

VISION MATE

Smart Assistant



PRESENTED BY

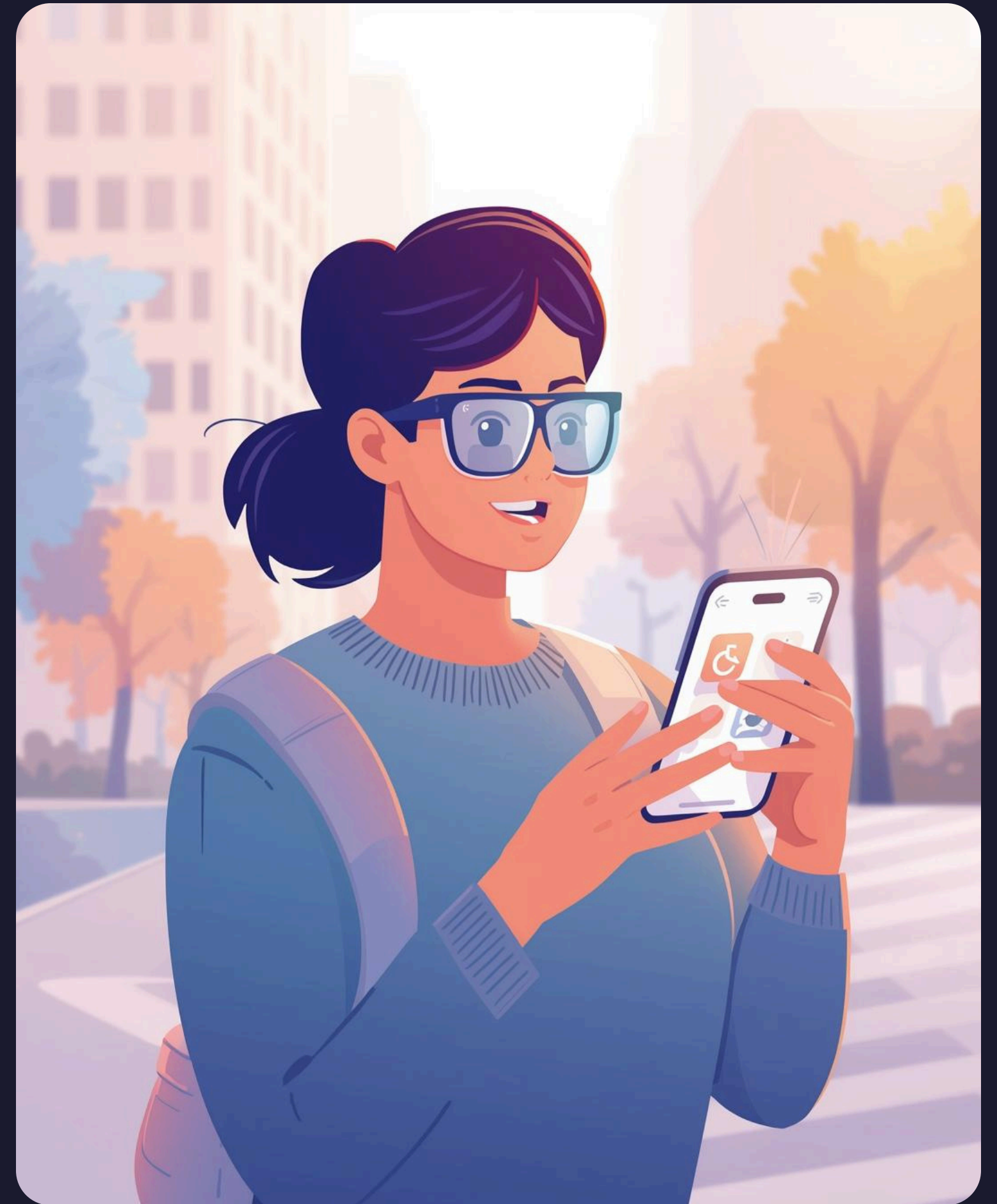
Samy Mahmoud

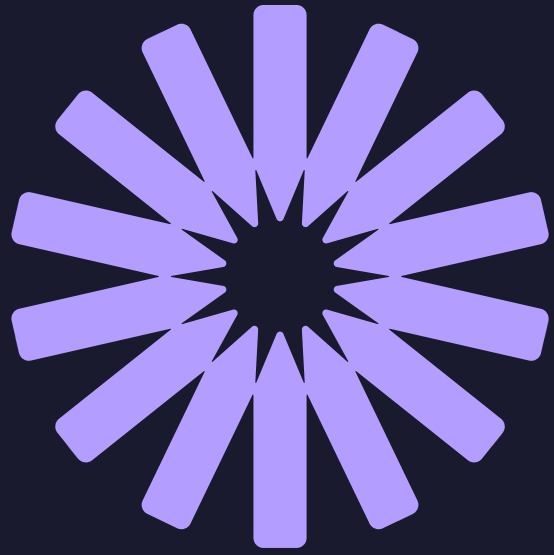
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Identifying Objects

Visually impaired individuals experience **significant challenges** identifying surrounding objects, impacting their ability to navigate safely.

Reading Challenges

Many visually impaired people struggle with reading printed text, hindering their access to information and independence.

The Problems Faced by the Visually Impaired

Project Objectives

Develop an Accessible Mobile Application

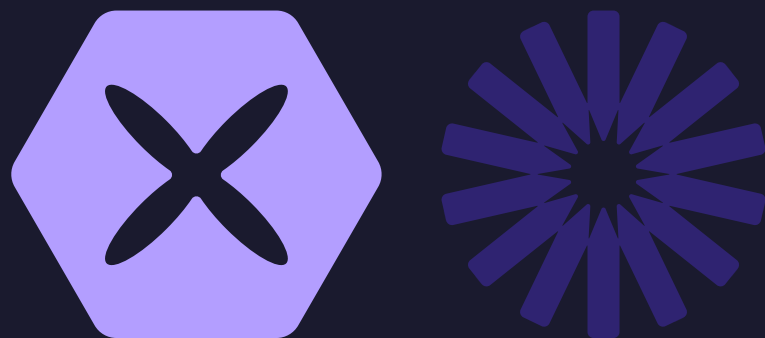
We aim to create a user-friendly mobile app that caters to the needs of visually impaired users, ensuring easy navigation and accessibility.

Achieve High AI Accuracy for Recognition

Our goal is to implement advanced AI algorithms that provide high accuracy in object and text recognition, empowering users with reliable information.

Provide Real-Time Voice Feedback

The app will deliver immediate voice feedback to users, facilitating prompt assistance during navigation and daily tasks, enhancing their independence and confidence.



Project Scope

The **project scope** defines the essential features that Vision Mate will deliver. In Scope includes critical aspects such as Object Detection, which allows users to identify their surroundings accurately. OCR technology enhances accessibility by converting printed text into digital format. The Text-to-Speech functionality ensures that users receive real-time audio feedback, while an Accessible User Interface design guarantees ease of use for visually impaired individuals.

Conversely, the **Out of Scope** elements emphasize boundaries for this project. Notably, the development of Hardware Devices is excluded, as the focus remains on software solutions. Advanced GPS navigation systems will also not be included, ensuring that resources are concentrated on refining the mobile app's features for object detection and text recognition. This clear distinction allows the team to maintain focus on delivering a functional and effective product.

Project Scope

In Scope:



Object Detection



OCR



Accessible UI design



UI/UX

Out of Scope:



Text-to-Speech



IoT

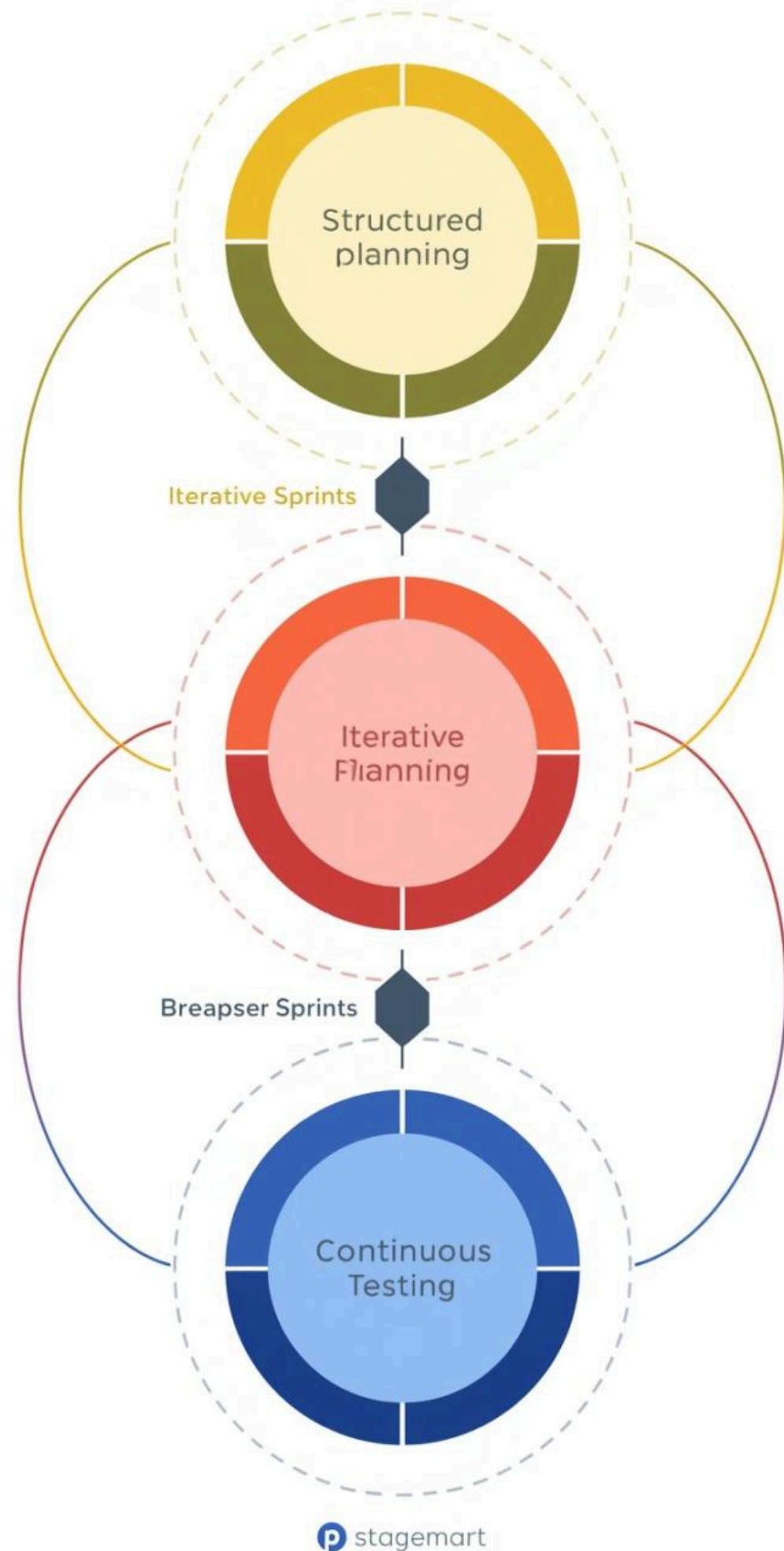


Advanced GPS



AI

Hybiect waistlingrer



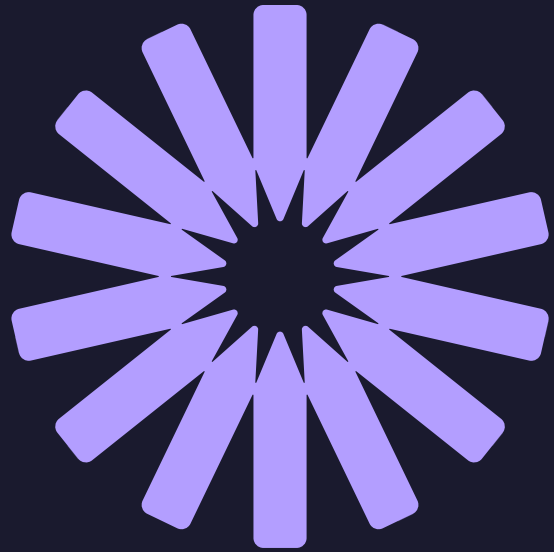
Hybrid Methodology

Combining Planning and Agile Development

This visual representation highlights the structured approach of **hybrid methodology**, blending traditional planning with agile development for optimal results.

Project Timeline





Project Manager

The Project Manager oversees all aspects, ensuring timely delivery, effective communication, and alignment with project goals.

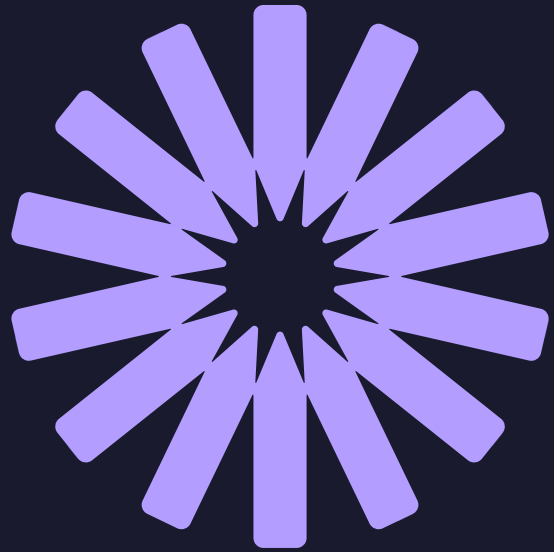
Flutter Developer

The Flutter Developer is responsible for crafting the mobile application, focusing on a responsive and intuitive user experience.

AI Integration Developer

The AI Integration Developer implements advanced AI features, including object and text recognition, enhancing app functionality.

Team Roles Overview



UI/UX Designer

The UI/UX Designer ensures that the app is accessible, intuitive, and user-friendly for visually impaired users.

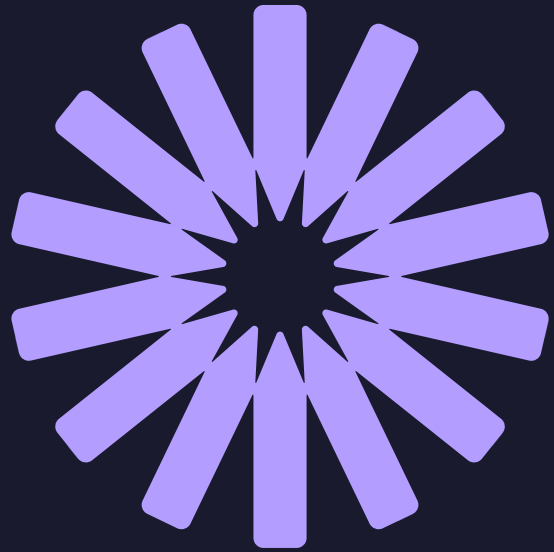
QA Engineer

The QA Engineer conducts thorough testing to ensure the app functions correctly and meets quality standards.

Collaboration

Effective collaboration among team members enhances creativity and leads to a more robust final product.

Team Roles and Responsibilities



Low AI Accuracy

To mitigate low accuracy, we will continuously **optimize and retrain** our AI models based on user feedback.

Performance Issues

We will conduct regular code **optimizations and profiling** to ensure our application runs smoothly across devices.

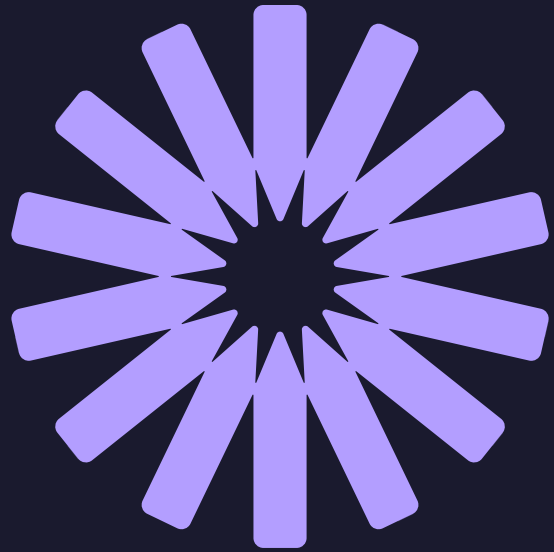
Team Delays

Weekly meetings will be scheduled to monitor **progress and address any issues** that may cause project delays.

Device Compatibility

Comprehensive testing on various devices will ensure that our app is **compatible** with different operating systems and screen sizes.

Risk Management Strategies Overview



Device Compatibility

Ensuring the app functions on various devices is crucial for reaching a wider audience and user satisfaction.

OS Variability

Compatibility across different operating systems ensures users can access the app regardless of their device choice.

Screen Sizes

Designing for multiple screen sizes enhances usability and accessibility for all users, accommodating different devices.

Continuous Testing

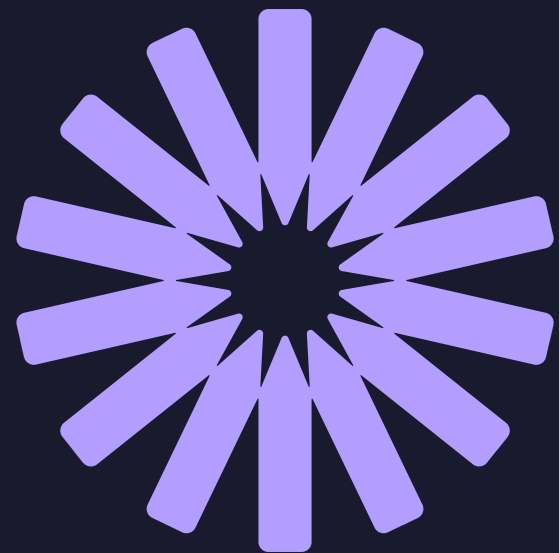
Regular testing on various devices helps identify and resolve compatibility issues before the app's full release.

Strategies for Device Compatibility

85%

Object Detection

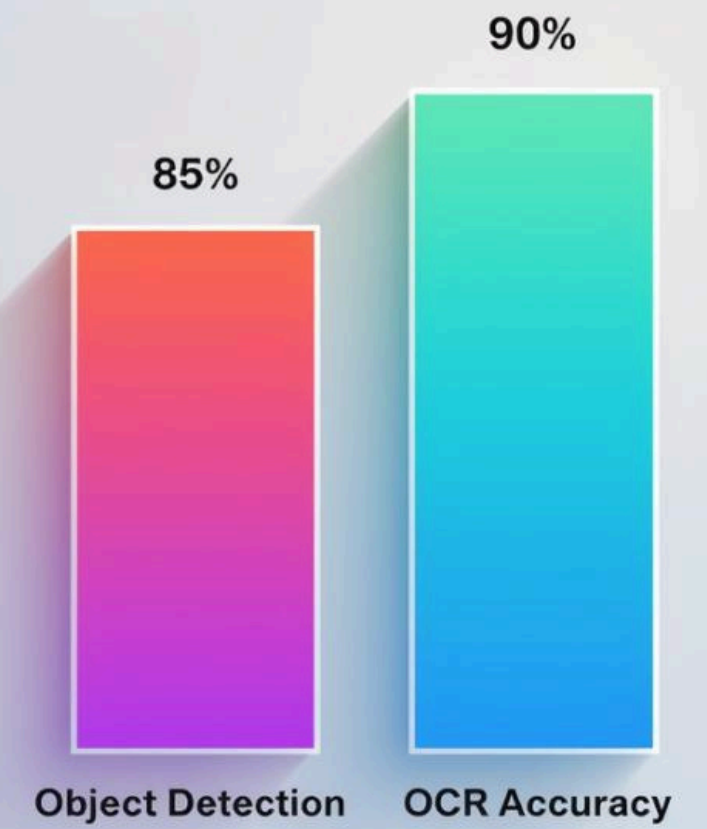
Achieving accuracy for identifying surrounding objects



90%

OCR Accuracy

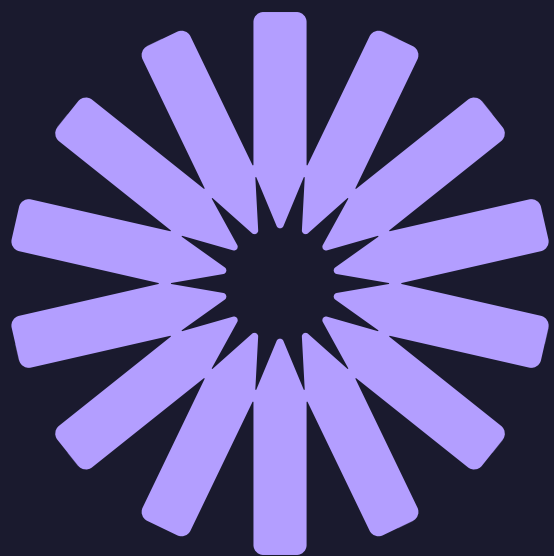
High accuracy for reading printed text



95%

Response Time

Users receive feedback within **two seconds** of action.



4%

Crash Rate

The application maintains an **impressive stability** record.





Impact of Vision Mate

Enhancing independence for visually impaired users

Vision Mate empowers users by improving safety, increasing accessibility, and fostering social inclusion through innovative technology.

Next Steps

Progressing Towards Our Goals

- Pilot testing with target users
- Collecting user feedback for improvements
- Planning for wider deployment
- Invitation to collaborate and support





“Technology is best when it
brings people together and
empowers their lives.”